



In everyday life, we often need to “count” the number of ways to arrange objects, or to perform a task, or to distribute items according to some specifications. For example, we may want to know the number of ways to divide a group of 13 people into three groups of 8 people, 4 people, and 1 person to occupy a van, a car, and a 2-seater convertible respectively.

Or if you went on a shopping spree and bought two pairs of jeans, three shirts, and two pairs of shoes, you may ask how many new outfits (consisting of a new pair of jeans, a new shirt, and a new pair of shoes) would you have.

The answers to questions like these can be obtained by listing all the possibilities, or using these counting techniques: the fundamental principles of counting, permutations and combinations. Collectively, these techniques are from the area of mathematics known as combinatorics.

Key Notation to Master: The Factorial

$$n! = 1 \cdot 2 \cdot 3 \cdots (n-2)(n-1)n, \text{ where } n \in \mathbb{Z}^+ \quad \text{and} \quad 0! = 1 \text{ (by definition).}$$

§1 Fundamental Principles of Counting

Key Question:

☐ When do we use the addition and multiplication principles?

1.1 Addition Principle

If there are r options for performing a particular task, and the number of ways to carry out the k^{th} option is n_k , for $k = 1, 2, 3, \dots, r$, then the total number of ways of performing the particular task is equal to the sum of the number of ways for all the r different options, i.e.,

$$n_1 + n_2 + n_3 + \cdots + n_r.$$

Note: The options to perform the task **cannot occur at the same time**.

When do we use the addition principle?

Example 1.1.1

An NJC student has the option of taking 3 different direct buses or taking a taxi to college. How many ways can she go to school?

We use the addition principle since to get to college (i.e. task), you EITHER take bus OR taxi.

Solution:

Mode (Option) of transport $\begin{cases} \text{Bus} \rightarrow 3 \text{ ways} \\ \text{Taxi} \rightarrow 1 \text{ way} \end{cases}$

Total number of ways = $3 + 1 = 4$.

When do we use the addition principle?

Example 1.1.2

A class consists of five girls (Amy, Betty, Carol, Denise, Elizabeth) and three boys (Charlie, Dan, Ethan). How many students are there in the class who is a girl or has a name starting with the letter 'E' or both?

Solution:

We can consider the problem as counting the total number of ways to perform the task of picking either a girl or a student whose name starts with the letter 'E'.

To use the addition principle, the options cannot occur at the same time, which is not the case here. So we cannot use it to get the answer directly.

Options	to	carry	out	task
{	Pick a girl			→ 5 ways
	Pick a student with name starting with 'E'			→ 2 ways

Note that "picking Elizabeth" is counted twice (i.e. picking a girl and a student with name starting with letter 'E'). Hence we need to subtract 1 from the sum of 5 and 2.

Hence the total number of students who is a girl or has a name starting with the letter 'E' = $5 + 2 - 1 = 6$.

Recall in Set Theory, we have

$$n(A \cup B) = n(A) + n(B) - n(A \cap B),$$

where $n(X)$ represents the number of elements in the set X (here X can be A , B , $A \cup B$ or $A \cap B$).

1.2 Multiplication Principle

If one task can be performed in m ways, and following this, a second task can be performed in n ways (regardless of which way the first task was performed), then the number of ways of performing the two tasks in succession is equal to product of the number of ways for both the tasks, i.e.,

$$m \times n.$$

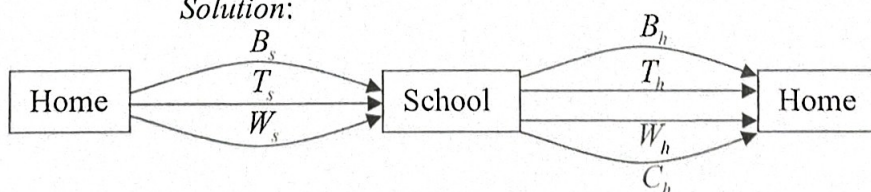
Note: This can be applied to two or more tasks performed **independently in succession**.

When do we use the multiplication principle?

Example 1.2.1

An NJC student has the option of taking either a bus (B), taxi (T) or walk (W) to school. For his return journey, he has the additional option of waiting for his dad to come fetch him in a car (C). How many different ways can he travel to school and back home?

Solution:



Task 1: Travel from home to school
There are 3 ways of doing this.

The student can travel to school and back home in the following ways:

$(B_s, B_h), (B_s, T_h), (B_s, W_h), (B_s, C_h),$

$(T_s, B_h), (T_s, T_h), (T_s, W_h), (T_s, C_h),$

$(W_s, B_h), (W_s, T_h), (W_s, W_h), (W_s, C_h).$

Task 2: Travel from school to home
There are 4 ways of doing this.

(For example, (B_s, W_h) corresponds to the student taking a bus to school and then walking home.)

Note that Task 1 and Task 2 are performed independently in succession so we may use the multiplication principle.

By the multiplication principle,
the total number of ways $= 3 \times 4 = 12$.

When do we use the addition and multiplication principles?

Example 1.2.2

An NJC student has the option of either taking a bus, walk or cycle for both his journey to school and his return journey home every day. However, he can only cycle home and must cycle home from school if he cycles to school on a particular day (*). How many different ways can he travel to college and back home on any day?

The tasks of travelling to school and travelling back home are not done independently (refer to *). So we cannot use the multiplication principle to get the answer directly. Instead, we consider 2 cases.

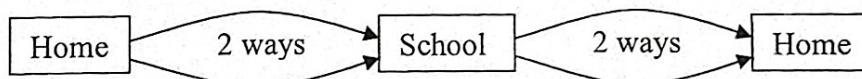
Solution:

We consider two different cases.

Case 1: Take the bus or walk to school.

In this choice, the student has two ways to travel home from school – either by bus or by walking, regardless of how he travelled to school.

Hence for case 1, by the multiplication principle, the total number of ways $= 2 \times 2 = 4$.



Case 2: Cycle to school.

In this case, the student has exactly 1 way to travel from home to school and vice versa, that is, to cycle for both journeys (as specified in the question).

The two cases cannot occur at the same time, i.e., cases are mutually exclusive. Hence addition principle is applicable.

Hence by addition principle, total number of ways $= 4 + 1 = 5$.

§2 Permutations

Key Questions:

- ☐ How do we find the number of permutations of r objects in a **ROW**
- from r **distinct** objects?
 - from r **objects**, some of which are identical?
 - from n ($\geq r$) **distinct** objects, where repetition is allowed?
- ☐ How do we find the number of permutations of n **DISTINCT** objects in a **CIRCLE**?

A permutation is an **ordered arrangement of objects**.

In permutations, order matters. For example, the 3-letter arrangement of ABC and ACB are considered as two different arrangements.

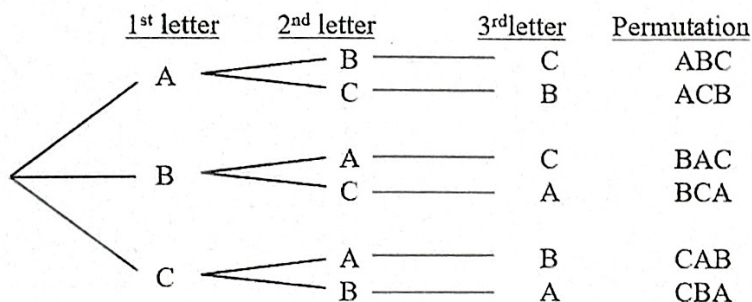
Activity 2.1

Which of the following is a permutation? Justify your responses.

1. Arrange three identical books on a shelf
2. Arrange three distinct books on a shelf
3. Select winners for the 1st, 2nd and 3rd prize
4. Arrange five people to sit in a row
5. Form a 3-digit number using the integers 1, 2, 3, 4, 5

2.1 Permutation of n DISTINCT objects in a ROW

Suppose we are interested to find the total number of possible 3-letter arrangements of A, B and C. We can do this by using a *tree diagram* as follows:



From the tree diagram, we see that the total number of possible 3-letter arrangements is 6. However, it would be very tedious and impractical to use a tree diagram to solve similar problems involving large number of letters, such as finding the total number of possible 5-letter arrangements of A, B, C, D, and E.

Alternatively, for the 3-letter arrangement, we can consider the problem as carrying out three separate tasks in succession – choosing the 1st, 2nd and 3rd letters respectively as follows:

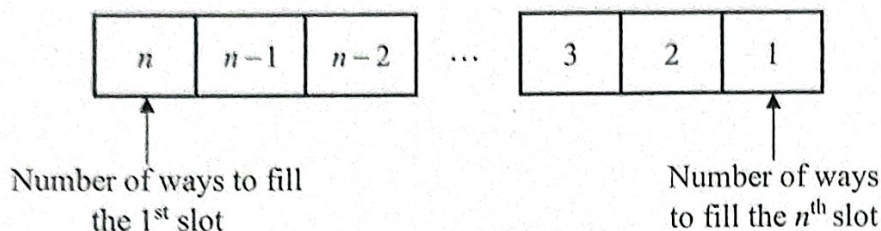
No. of ways to pick the 1 st letter	No. of ways to pick the 2 nd letter	No. of ways to pick the 3 rd letter
3	2	1

Regardless of the 1st letter, there are always only 2 ways to pick the 2nd letter. Why?

Hence by the multiplication principle,
the total number of possible 3-letter arrangements of A, B and C = $3 \times 2 \times 1 = 3! = 6$.

We can generalise the above problem to obtain the following result.

Given n distinct objects, the total number of ways of **arranging all these n objects in a row** can be deduced as:



Since we consider filling the slots from left to right as tasks done independently in succession, we apply the multiplication principle here.
Hence, the total number of ways is

$$n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 = n!.$$

Note: This method of filling “one slot at a time” takes order into consideration.



WONDER

How many ways are there to arrange 0 objects?

Answer: One. This is one of the several ways to interpret the definition of $0!$

How to arrange n distinct objects in a row?

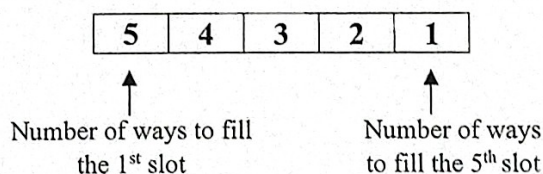
Example 2.1.1

Isabel has five different books. How many ways can she arrange all five books on a shelf?

Solution:

The order (i.e. which book comes first) matters.

The method of “filling the slots” to count is more efficient than listing all the possible permutations.



Therefore number of ways = $5! = 120$.

2.2 Permutation of r objects taken from n DISTINCT objects in a ROW

How to arrange r objects in a row taken from n distinct objects? **Example 2.2.1**

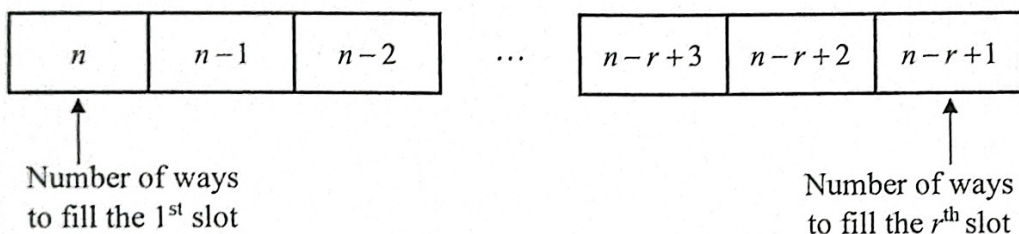
Isabel has five different books. How many ways can she arrange three books on a shelf?

Consider the method of filling the r slots. **Solution:**

Number of ways $= 5 \times 4 \times 3 = 60$.

We can generalise the above problem to obtain the following result.

Given n distinct objects, the number of ways arranging r of these objects ($1 \leq r \leq n$) in a row is:



Hence, the total number of ways

$$\begin{aligned}
 &= n(n-1)(n-2) \dots (n-r+2)(n-r+1) \\
 &= n(n-1)(n-2) \dots (n-r+2)(n-r+1) \cdot \frac{(n-r)(n-r-1) \dots 3 \cdot 2 \cdot 1}{(n-r)(n-r-1) \dots 3 \cdot 2 \cdot 1} \\
 &= \frac{n!}{(n-r)!} \\
 &= {}^n P_r
 \end{aligned}$$

Remarks:

- For convenience, we extend the formula to include $r=0$, so that

$${}^n P_0 = \frac{n!}{(n-0)!} = 1.$$

- When $r=n$, ${}^n P_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$. (Recall that $0! = 1$.)

How to arrange r objects in a row taken from n distinct objects? **Example 2.2.2**

How many 6-letter code words (no repetition of letters) can be formed from the letters of the word "SINGAPORE"?

Solution:

Notice that we do not allow repetition of letters in the 6-letter code words and that there are nine distinct letters in the word "SINGAPORE".

Number of 6-letter code words

$$= {}^9P_6 = \frac{9!}{(9-6)!} = 60,480.$$

Alternatively, we consider filling the 6 slots.

9 8 7 6 5 4



Number of ways
to fill the 1st slot



Number of ways
to fill the 6th slot

Number of 6-letter code words

$$= 9 \times 8 \times 7 \times 6 \times 5 \times 4 = 60,480.$$

How to arrange r objects in a row taken from n distinct objects?

Example 2.2.3

How many 4-digit numbers can be formed using the digits 1, 2, 3, 5, 7,

- if no repetition of the numbers is allowed?
- if repetition is allowed?

Solution:

- Consider filling the four slots using distinct digits

5 4 3 2

Number of 4-digit numbers

$$= 5 \times 4 \times 3 \times 2 \text{ (or } {}^5P_4) = 120.$$

Compare how we fill in the slots when we allow repetition (versus do not allow repetition).

- 5 5 5 5



Number of ways
to fill the 1st slot



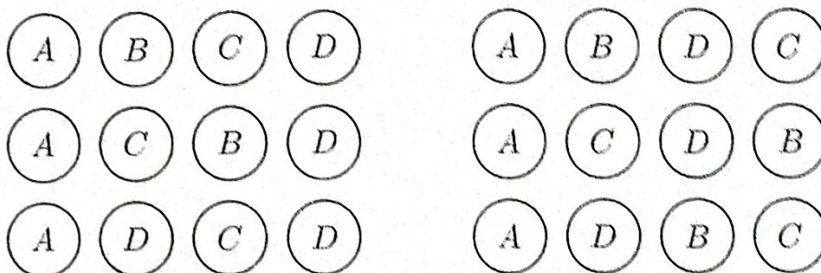
Number of ways
to fill the 4th slot

$$\text{Number of 4-digit numbers} = 5^4 = 625.$$

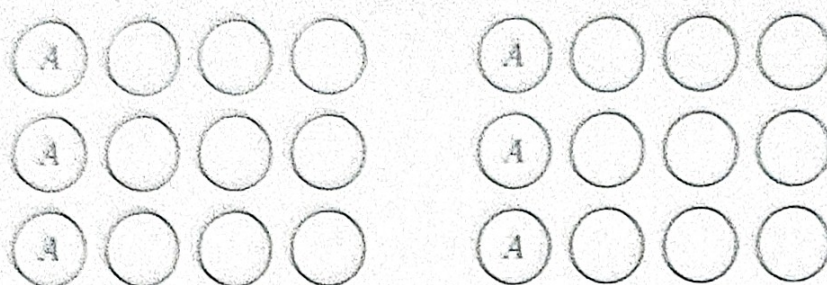
2.3 Permutation of n objects, SOME of which are IDENTICAL, in a ROW

Consider the arrangement of four balls labelled as A, B, C, D in a straight line. The total number of ways to arrange these four *distinct* objects = $4!$.

For the case where A is the first ball, there are a total of $3!$ or 6 possible arrangements (since there are $3!$ ways of permuting the three remaining balls) as follows:



Now suppose that the balls labelled B, C and D are replaced by *indistinguishable* unlabelled balls (i.e. considered as identical).



Now the 6 (or $3!$) originally different possible arrangements are considered to be the same arrangement. This occurs for the cases where A is the 2nd, 3rd or 4th ball as well. Hence, the total number of ways to arrange four objects with three being identical (or indistinguishable) = $\frac{4!}{3!}$.

The number of ways to arrange n objects, of which exactly m objects are identical to one another, in a row is $\frac{n!}{m!}$.

We also have the following general result.

Suppose there are n objects, of which there are r different groups of identical / non-distinct / indistinguishable objects. The total number of permutations of all the n objects in a row, is given by

$$\frac{n!}{n_1! n_2! \dots n_r!}$$

where n_k represents the number of identical objects in the k^{th} group, for $k = 1, 2, 3, \dots, r$.

Note: The division by $n_1! n_2! \dots n_r!$ is to compensate for the over counting under the assumption that the objects in each of the r groups are distinct.

How to arrange n objects in a row, of which some are identical objects? **Example 2.3.1**

How many ways can we arrange two identical white balls and three identical red balls in a row?

Solution:

Groups of identical objects		
Number of identical objects in each group	2	3

Since there are five balls, total number of ways to arrange the balls = $\frac{5!}{2!3!} = 10$.

The 10 possibilities are:

WWRRR RWWRR RRWWR RRRWW WRWRR
RWRWR RRWRW WRRWR RWRRW WRRRW

How to arrange n objects in a row, of which some are identical objects?

Example 2.3.2

How many ways can we arrange the letters of the word PARAGRAPH in a row?

Solution:

We first tally the number of times each letter appears in the word PARAGRAPH.

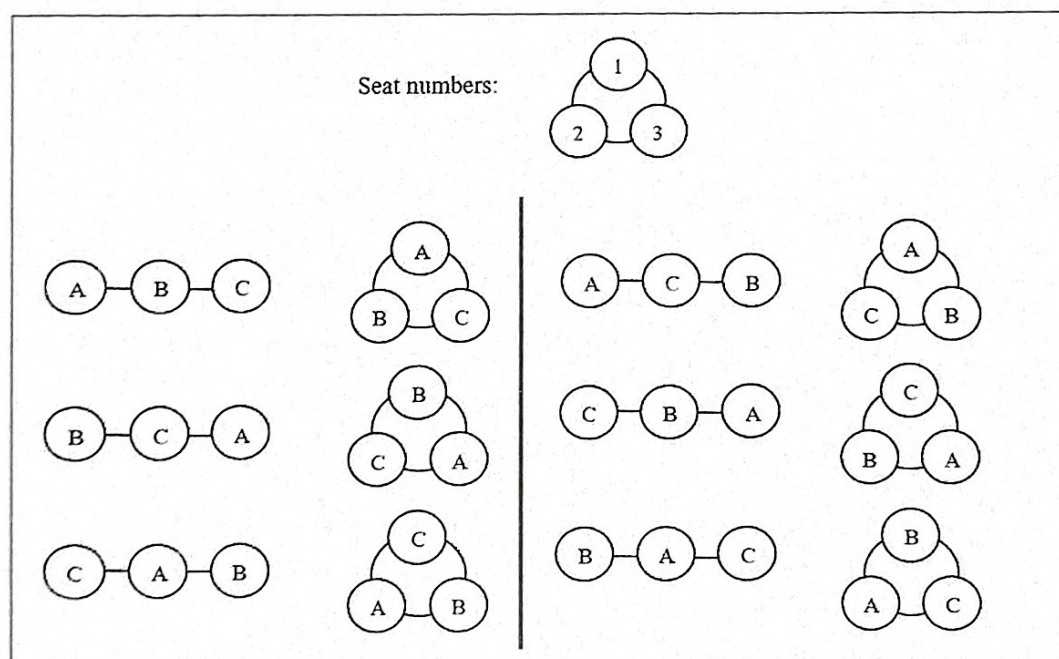
P	A	R	G	H
2	3	2	1	1

$$\text{Number of ways} = \frac{9!}{2!2!3!} = 15,120.$$

2.4 Permutation of n DISTINCT objects in a CIRCLE

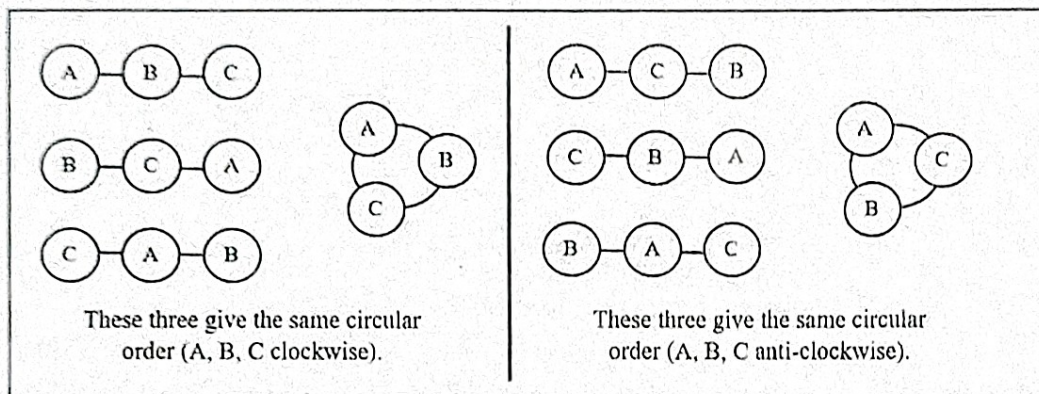
In a row, there is the notion of “first position” and “last position”. For example, when arranging persons A, B and C, the three row arrangements ABC, BCA, and CAB are different. Altogether, there are $3! = 6$ different ways to arrange three persons A, B, and C in a row.

Notice what happens when we join the ends of each row to form a circle, where the seats are numbered from 1 to 3 in the following manner:



If the seats are numbered, then all the 6 circular arrangements of A, B, and C are distinct, and it is actually a row problem (in disguise).

On the other hand, if the seats are not numbered, then there is no first or last position to consider. Hence the three row arrangements ABC, BCA, and CAB are really duplicates and can be classified as one same circular arrangement.



Note however that **clockwise and anticlockwise circular arrangements are different**. (To see this, imagine you are A. Then the person sitting on your left hand side is different for the two circular arrangements).

Another way of thinking is that ABC, BCA, and CAB is really the same circular arrangement but rotated anti-clockwise. So, by using the formula for the total number of ways to permute three distinct objects in a row to find the total number of possible arrangements in a circle, we would have over-counted by a factor of 3. Hence, to obtain the correct number, we divide $3!$ by 3.

This reasoning can be extended to circular arrangements of more objects.

The number of permutations of n distinct objects in a circle (where the seats are indistinguishable) is given by $\frac{n!}{n} = (n-1)!$.

Note: Clockwise and anticlockwise circular arrangements are different.

Remark: Unless specified otherwise, all seats are not numbered when circular arrangement is involved.

How to arrange n distinct objects in a circle?

Example 2.4.1

In how many ways can five men be seated at a round table?

Solution:

$$\text{Number of ways} = \frac{5!}{5} = (5-1)! = 4! = 24.$$

How to arrange n distinct objects in a circle?

Example 2.4.2

How many ways can four friends use their left hands to hold another friend as shown in the diagram?



Reframe this problem as arranging the 4 friends in a circle.

Solution:

$$\text{Number of ways} = \frac{4!}{4} = (4-1)! = 3! = 6.$$

§3 Combinations

Key Questions:

- ☐ How do we find the number of selections of r objects from n **distinct** objects?
- ☐ What is the difference between permutations and combinations?
 - What is an equation relating ${}^n P_r$ with ${}^n C_r$?
 - Interpret the relationship between ${}^n P_r$ and ${}^n C_r$.

A combination is a **selection of objects** in which the **order of selection does not matter** (unlike in permutations).

For example, in a selection of two students from a class (say, Anthony and Belle), it does not matter if Anthony or Belle is selected first.

Activity 3.1

Which of the following is a combination? Justify your responses.

- Choosing a committee of three people from a group of ten people.
- Picking a set of three integers from the numbers 1, 2, 3, 4, 5.
- From a box of 200 lucky draw tickets, three tickets are to be drawn for three different consolation prizes.

3.1 Combination (Selection) of r objects taken from n DISTINCT objects

The number of ways to select r objects from n **distinct** objects is denoted by $\binom{n}{r}$ (or ${}^n C_r$) where

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad r \leq n.$$

(Why is $r \leq n$?)

Note: When using $\binom{n}{r}$, the n objects **MUST BE DISTINCT**.

How to select r objects from n ($\geq r$) distinct objects?

Example 3.1.1

A club consists of 3 doctors, 9 teachers and 5 lawyers. Find the number of ways a three-member team can be chosen from the club if

- there is no restriction,
- it comprises 1 doctor and 2 lawyers.

Solution:

- There are a total of $3+9+5=17$ people in the club.

$$\text{Number of ways} = \binom{17}{3} = 680.$$

$$(ii) \text{ Number of ways to choose 1 doctor} = \binom{3}{1}.$$

The selection of 1 doctor and 2 lawyers are 2 independent tasks performed in succession. Hence we apply the multiplication principle.

$$\text{Number of ways to choose 2 lawyers} = \binom{5}{2}.$$

$$\text{Therefore, number of ways} = \binom{3}{1} \binom{5}{2} = 30.$$

Special Results	Interpretation
$\binom{n}{0} = 1$	There is only 1 way to choose 0 object from n distinct objects, i.e. take none.
$\binom{n}{n} = 1$	There is only 1 way to choose n objects from n distinct objects, i.e. take all.
$\binom{n}{1} = n$	There are n ways to choose 1 object from n distinct objects, i.e. take any 1 of the n objects.
$\binom{n}{r} = \binom{n}{n-r}$	If we choose r objects from n distinct objects, there would be $(n-r)$ remaining objects, and vice versa. Hence the number of ways to select r objects out of n distinct objects is <u>equal to</u> the number of ways to select $(n-r)$ objects out of n distinct objects.
$\binom{n}{n-1} = \binom{n}{1} = n$	The number of ways to choose $(n-1)$ objects from n distinct objects is equal to the number of ways to choose 1 object from n distinct objects (from above result). Algebraically, we have $\binom{n}{n-1} = \frac{n!}{(n-1)!(n-(n-1))!} = \frac{n!}{(n-1)!1!} = \frac{n(n-1)!}{(n-1)!} = n.$
$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$	Refer to Appendix B for the proof.

3.2 Relationship between Permutations and Combinations

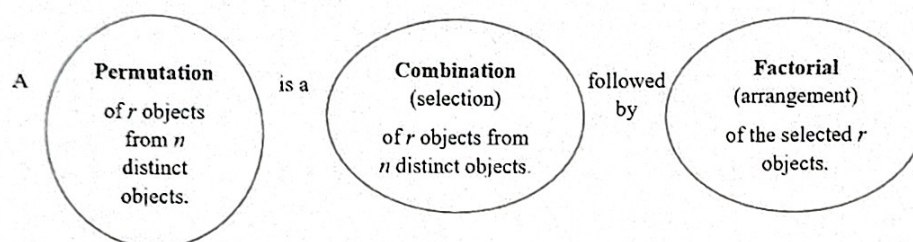
Recall that

- a permutation is an ordered **arrangement** of objects, and
- a combination is an unordered **selection** of objects.

Algebraically, since ${}^nP_r = \frac{n!}{(n-r)!}$ and ${}^nC_r = \frac{n!}{r!(n-r)!}$, we have

$${}^nP_r = \frac{n!}{(n-r)!} = \frac{n!}{r!(n-r)!} \times r! = {}^nC_r \times r!. \quad \text{--- (*)}$$

We may interpret (*) as



Note: ${}^nP_r = {}^nC_r \times r! = \binom{n}{r} \times r!$

What is the **Example 3.2.1**

relationship between permutation and combination? How many ways can we arrange three eggs from a basket of nine different Easter eggs?

Method 1 considers **Solution:**

filling of the three slots as illustrated in Section 2.2. Method 1: Number of ways = ${}^9P_3 = 9 \times 8 \times 7 = 504$.

Method 2 considers selection of three distinct eggs followed by arrangement of the three eggs.

Method 2: Number of ways = $\binom{9}{3} \times 3! = 504$.

§4 Problem-Solving Strategies in Permutations & Combinations

Key Questions:

- ☐ What are some strategies to solve problems including cases involving repetitions and restrictions?

We follow the *general rule of selecting first then arranging* (since we need to select objects before we can arrange them).

When restriction(s) are imposed, we apply the *general rule of dealing with the restriction(s) first before dealing with the rest*.

How is the general rule applied? Select then arrange!

Since Tasks 1 and 2 are performed independently in succession, we apply the multiplication principle.

Example 4.1

Find the number of ways to choose a captain and a vice-captain from nine students.

Solution:

Task 1: **Choose** two students from the nine students

Number of ways = 9C_2

Task 2: **Decide** who will be the captain and vice-captain

Number of ways = $2!$

Thus, the total number of ways is ${}^9C_2 \times 2! = 72$.

Strategy:

Considering cases
(since no single direct calculation suffices)

Since there could be different number of black/white balls in the five selected from the box and different compositions of the black/white balls warrant slightly different calculations for the number of arrangements of the five balls, we need to consider cases to fix the compositions of black/white balls to proceed.

Example 4.2

A box contains four identical black balls and three identical white balls. Find the number of ways five balls from the box can be arranged in a row.

Solution:

Consider cases to fix the number of identical balls in the five balls selected.

Case 1: One white and four black balls.

Note that there is only 1 way to select one white and four black balls.

Number of ways = $\frac{5!}{4!} = 5$.

Case 2: Two white and three black balls.

Note that there is only 1 way to select two white and three black balls.

Number of ways = $\frac{5!}{3!2!} = 10$.

Case 3: Three white and two black balls.

Note that there is only 1 way to select three white and two black balls.

Since the cases do not occur at the same time, we apply the addition principle.

$$\text{Number of ways} = \frac{5!}{2!3!} = 10.$$

$$\text{Total number of ways} = 10 + 10 + 5 = 25.$$

A student proposes that we can use $\frac{{}^7P_5}{3!4!}$ to calculate the total number of ways directly. Why is this not a valid method?

Strategy:

Considering cases
(since no single direct calculation suffices)

Since there are two possible vowels to be placed as the 1st letter and for each choice of vowel placed as the 1st letter, there will be different numbers of identical letters left to be arranged, which warrant different calculations for the number of arrangements. Hence we need to consider cases to fix the vowel to be placed as the 1st letter.

Since the cases do not occur at the same time, we apply the addition principle.

Example 4.3

Find the number of ways the letters of “INDEPENDENT” can be arranged in a row such that the first letter must be a vowel.

Solution:

I	N	D	E	P	T
1	3	2	3	1	1

There are 2 cases for the 1st letter to be a vowel.

Case 1: 1st letter is I.

The remaining ten letters have three identical N, two identical D and three identical E.

$$\text{Number of ways} = \frac{10!}{3!2!3!} = 50,400.$$

Case 2: 1st letter is E.

The remaining ten letters have three identical N, two identical D and two identical E.

$$\text{Number of ways} = \frac{10!}{3!2!2!} = 151,200.$$

$$\text{Total number of ways} = 50,400 + 151,200 = 201,600.$$

Strategy:

Considering cases
(since no single direct calculation suffices)

As highlighted in the previous two examples, we consider cases when there is NO single calculation that gives the answer directly since the calculations vary with different scenarios (which will be framed as cases).

Example 4.4

A class of 30 students comprises 13 boys and 17 girls. Five students are to be selected to form a class committee. Find the number of ways such that the class committee contains at least three girls and at least one boy.

Solution:

Case 1: Exactly three girls and two boys

$$\text{Number of ways} = \binom{17}{3} \binom{13}{2} = 53040$$

Case 2: Exactly four girls and one boy

$$\text{Number of ways} = \binom{17}{4} \binom{13}{1} = 30940$$

$$\text{Total number of ways} = 53040 + 30940 = 83980.$$

Note: In each case, we apply multiplication principle since there are two tasks, namely selecting girls and selecting boys, to be performed independently in succession.

Strategy:

Taking the complement (in order to consider less cases)

Example 4.5

A class of 30 students comprises 13 boys and 17 girls. Five students are to be selected to form a class committee. Find the number of ways such that the class committee contains at least one girl.

Solution:

Number of ways

= Total number of ways – Number of ways where no girl is selected (i.e. five boys selected)

$$= \binom{30}{5} - \binom{13}{5}$$

$$= 141219.$$

An alternative (less efficient) method is to consider 5 different cases:

Case 1: Exactly 1 girl (and 4 boys)

Case 2: Exactly 2 girls (and 3 boys)

Case 3: Exactly 3 girls (and 2 boy)

Case 4: Exactly 4 girls (and 1 boy)

Case 5: Exactly 5 girls (and no boy)

Strategy:

Bundling (since objects need to be placed **TOGETHER**)

Example 4.6

Find the number of ways four boys and three girls can be arranged in a row such that the three girls are next to one another.

Solution:

“Bundle” the girls who have to be together as 1 group.

$\boxed{G_1 G_2 G_3} B_1 B_2 B_3 B_4$

Since Tasks 1 and 2 are performed independently in succession, we apply the multiplication principle.

Task 1: Arrange four boys and one group of girls

Number of ways = $5!$.

Task 2: Arrange the three girls within the “bundle”

Number of ways = $3!$.

Number of ways = $5!3! = 720$.

Strategy:

Slotting-in (since objects need to be **SEPARATED**)

Example 4.7

Find the number of ways four boys and three girls can be arranged in a row if all the girls are separated.

Solution:

We arrange those with no restrictions first, before slotting those that need to be separated.

Note that this is contrary to the general rule of dealing with the restriction(s) first before dealing with the rest.

Arrange the boys first

↑ B ↑ B ↑ B ↑ B ↑

Possible slots for girls

Task 1: Arrange the boys in a row

Number of ways = $4!$

Task 2: Slot in the three girls among the five spaces between the boys (i.e. choose 3 spaces and arrange the girls in these three spaces)

Number of ways = $5 \times 4 \times 3 = 60$

Total number of ways = $60 \times 4! = 1440$

Since Tasks 1 and 2 are performed independently in succession, we apply the multiplication principle.

Can we use the complement method to compute the number of arrangements where the three girls must be separated?

Strategy:

Alternate objects (this is NOT to be confused with "separated")

Notice that we consider cases as the first person in the row can be a boy or girl.

Depending on the number of boys and girls, we may not necessarily need to consider cases. For instance, if we have four boys and five girls, then there is only 1 way to alternate the boys and girls, i.e. GBGBGBGBG.

Since the cases do not occur at the same time, we apply the addition principle.

Example 4.8

Find the number of ways five boys and five girls can be arranged in a row such that the boys and girls alternate.

Solution:

Case 1: The first person is a boy.

B G B G B G B G B G

Number of ways = $5! \times 5! = 14,400$.

Case 2: The first person is a girl.

G B G B G B G B G B

Number of ways = $5! \times 5! = 14,400$.

Hence total number of ways
 $= 14,400 + 14,400$
 $= 28,800$.

Why can't we apply the Slotting-In method in Example 4.7?

Strategy:

Taking the complement for "not all"

The keyword in this question is "not". Hence "not all girls are together" is read as NOT "all girls are together", which means we DO NOT WANT any arrangements where "all girls are together". This implies we take the complement of "all girls are together".

Example 4.9 ("not all" versus "all not")

- (a) Find the number of ways four boys and three girls can be arranged in a row if *not all* the girls are together.
- (b) Find the number of ways four boys and three girls can be arranged in a row if the girls are *all not* together.

Solution:

- (a) Number of ways to arrange four boys and three girls in a row = $7!$.

Since for any arrangement,
 EITHER all the 3 girls are together
 OR *not all* the 3 girls are together,

Therefore number of ways
 $= 7! - 720$ (answer in Example 4.6)
 $= 4320$

- (b) "girls are all not together" is equivalent to "girls are separated"
 Hence number of ways
 $= 1440$ (answer in Example 4.7)



WONDER

How are **Example 4.7** and **Example 4.9(a)** different?

Answer: **Example 4.7** calculates the number of ways where all the girls are separated, whereas **Example 4.9(a)** calculates the number of ways where *not all* the girls are together. In **Example 4.9(a)**, we allow arrangements where two girls are together but such arrangements are not allowed in **Example 4.7**.

Strategy:

Bundling and slotting-in applied in circular arrangement

Example 4.10

Four boys and three girls are to be seated at a round table. Adrian, Jack and Benjamin are three particular boys among the seven. How many arrangements are there if

- there is no restriction?
- Adrian, Jack and Benjamin must be together?
- Adrian, Jack and Benjamin must be separated?
- Selina must sit between Adrian and Benjamin?

How many arrangements are there if there is no restriction, but the seats are numbered?

Solution:

(i) Number of arrangements = $\frac{7!}{7} = 720$.

- (ii) “Bundle” Adrian, Jack and Benjamin together as 1 group first

Task 1: Arrange the group with the remaining four people around the table

Number of ways = $\frac{5!}{5}$.

Task 2: Arrange Adrian, Jack and Benjamin within the group

Number of ways = $3!$.

Hence number of arrangements = $\frac{5!}{5} \times 3! = 144$.

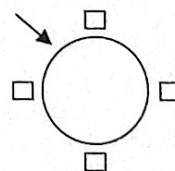
- (iii) Task 1: Arrange the other 4 people round the table

Number of ways = $\frac{4!}{4} = 3!$.

Task 2: Slot in Adrian, Jack and Benjamin among the 4 spaces between the four round the table (i.e. choose 3 spaces and arrange Adrian, Jack and Benjamin in these three spaces)

Number of ways

= $\binom{4}{3} \times 3!$ (or $4 \times 3 \times 2$) = 24.



Total number of ways = $3! \times 24 = 144$.

We apply the general rule of dealing with the restriction(s) first before dealing with the rest.

(iv) "Bundle" A S B as 1 group.

Task 1: Arrange the group with the remaining four people around the table

$$\text{Number of ways} = \frac{5!}{5} = 4!$$

Task 2: Arrange Adrian and Benjamin within the group

$$\text{Number of ways} = 2!$$

$$\text{Number of arrangements} = 4! \times 2! = 48.$$

If seats are numbered, it is equivalent to arranging them in a row.

$$\text{Therefore, number of arrangements} = 7! = 5,040.$$



WONDER

In (iii), why do we NOT divide by $3!$ when we arrange Adrian, Jack and Benjamin among the three spaces even though it is arranging round the table?

Answer: After arranging the other four people, the slots between each pair of adjacent persons are distinct i.e. we are now arranging in a row rather than in a circle.



WONDER

Why is the answer for (iii) not equal to answer for (i) – answer for (ii)?

Answer: If we take answer for (i) – answer for (ii), we will have arrangements where any two of Adrian, Jack and Benjamin are together, which are not allowed in (iii).

Strategy:

Distinct and Identical Groupings

Example 4.11

Calculate the number of ways in which 6 people can be divided into

- a group of 2 and a group of 4,
- 3 groups comprising 2 people each, where each group is working on a different project,
- three groups of 2,
- two groups of 2 and two groups of 1.

(i) Consider filling 2 distinct slots – 1 slot for two persons and the other for four persons. Task 1 is to select two persons from the six and Task 2 is to select four from

Solution:

$$(i) \text{ Number of ways} = \binom{6}{2} \cdot \binom{4}{4} = 15.$$

$$(ii) \text{ Number of ways} = \binom{6}{2} \cdot \binom{4}{2} \cdot \binom{2}{2} = 90.$$

the remaining four. Since both tasks are done independently in succession, we apply the multiplication principle.

The subsequent parts can also be explained using "filling in the slots" method.

(iii) The slots are identical hence the need to compensate for the over-counting by dividing by $3!$.

(iv) There are 2 identical slots (for 2 persons) and 2 identical slots (for 1 person). Hence we need to compensate for the over-counting by dividing by $2!2!$.

$$(iii) \text{ Number of ways} = \frac{\binom{6}{2} \cdot \binom{4}{2} \cdot \binom{2}{2}}{3!} = 15.$$

(iii) Possible ways to divide 6 people, say A, B, C, D, E, and F, into three groups of two each:

(AB) with (CD)(EF) (AE) with (BC)(DF)
(CE)(DF) (BD)(CF)
(CF)(DE) (BF)(CD)

(AC) with (BD)(EF) (AF) with (BC)(DE)
(BE)(DF) (BD)(CE)
(BF)(DE) (BE)(CD)

(AD) with (BC)(EF)
(BE)(CF)
(BF)(CE) Total: 15 ways.

$$(iv) \text{ Number of ways} = \frac{\binom{6}{2} \cdot \binom{4}{2} \cdot \binom{2}{1} \cdot \binom{1}{1}}{2! \cdot 2!} = 45.$$

Remark: When we choose $n (\geq 2)$ indistinguishable groups of the same size, each combination is repeated $n!$ times, NOT n times.

Strategy:

Reframing the problem

Example 4.12

A restaurant offers butter, cheese, chives, and sour cream as toppings for a baked potato. How many different ways are there to order a baked potato?

Solution:

From the perspective of the customer, the customer can choose to have 0 topping or 1 topping or 2 toppings or 3 toppings or 4 toppings. Thus,

$$\text{number of ways} = \binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 16.$$

Alternatively, from the perspective of the toppings, each topping can either be chosen or not chosen. Thus, number of ways $= 2^4 = 16$.

Strategy:
Reframing the
problem

We may consider the "fill-in-the-slots" method where each distinct slot represents a fruit.

Example 4.13

Find the number of ways for a boy to select fruit(s) from a basket of four oranges, six apples, two bananas, assuming the boy cannot be empty handed.

Solution:

The boy can select 0, 1, 2, 3 or 4 oranges.

So, the number of ways to choose oranges is 5.

The boy can select 0, 1, 2, 3, 4, 5 or 6 apples.

So, the number of ways to choose apples is 7.

The boy can select 0, 1 or 2 bananas.

So, the number of ways to choose bananas is 3.

$$\text{Number of ways} = (5 \times 7 \times 3) - 1 = 104$$

↑
Exclude the case where the boy selects 0 for all three fruits.

The table below summarises the strategies (non-exhaustive) used in the examples illustrated. More often than not, questions may have several restrictions that require more than one strategy.

Strategies:	When to use?
Consider cases	When there is NO single direct calculation to calculate the number of ways
Taking the Complement	In order to consider less number of cases
Bundling	When objects need to be placed TOGETHER
Slotting-in	When objects need to be SEPARATED
Alternate Objects	When two types of objects need to be in ALTERNATING positions E.g. boys and girls alternate. <i>This is NOT to be confused with "separated".</i>
Distinct and Identical Groupings	When distinct objects are to be placed in GROUPS <i>Need to be able to tell if the groups are distinct or identical from the context.</i>
Reframing the Problem	When question is not phrased as a selection or arrangement problem where we can apply the permutations or combination results introduced in Sections 2 and 3 directly. <i>Usually we need to take perspectives (as in Example 4.12) or use "fill in the slots" method (as in Example 4.13).</i>

Note that these strategies can be applied (with caution) either in **ROW** or **CIRCULAR** arrangements.

Appendix A: Use of GC

The following operations found in the Math PROB menu are very useful in evaluating your solutions.

Press **MATH** **)** **)** **)** to get to the menu.

NORMAL FLOAT AUTO REAL DEGREE MP			
MATH	NUM	CMPLX	PROB FRAC
1:rand			
2:nPr			
3:nCr			
4:!			
5:randInt(			
6:randNorm(			
7:randBin(			
8:randIntNoRep(			

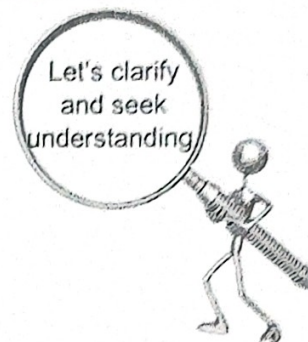
Operation:	Example:
2: nPr	To evaluate 5P_2 , $\begin{array}{r} {}^5P_2 \\ \hline 20 \end{array}$
3: nCr	To evaluate 5C_2 or $\binom{5}{2}$, $\begin{array}{r} {}^5C_2 \\ \hline 10 \end{array}$
4: !	To evaluate $3!$, $\begin{array}{r} 3! \\ \hline 6 \end{array}$

Appendix B: Proof of the result $\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$.

We present two methods to prove the above result.

Algebraic Proof

$$\begin{aligned}
 \text{RHS} &= \binom{n}{r} + \binom{n}{r-1} \\
 &= \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)![n-(r-1)]!} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} \\
 &= \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)(n-r)!} = \frac{n!(n-r+1) + n!r}{r!(n-r+1)!} \\
 &= \frac{n!n - n!r + n!r}{r!(n-r+1)!} = \frac{n!(n+1)}{r!(n+1-r)!} = \frac{(n+1)!}{r![(n+1)-r]!} = \binom{n+1}{r} = \text{LHS}.
 \end{aligned}$$



Combinatorial Proof (i.e. by using an argument involving permutations and combinations)

To choose r objects from $n+1$ distinct objects, we have 2 choices:

Choice 1: Choose all r objects from the first n objects.

$$\text{Number of ways to carry out choice 1} = \binom{n}{r}.$$

Choice 2: Choose the $(n+1)^{\text{th}}$ object and another $(r-1)$ objects from the first n objects.

We can view this as performing two tasks in succession:

- (1) Number of ways to choose the $(n+1)^{\text{th}}$ object = 1.
- (2) Number of ways to choose $(r-1)$ objects from the first n objects = $\binom{n}{r-1}$.

By the multiplication principle, number of ways to carry out Choice 2 is

$$1 \times \binom{n}{r-1} = \binom{n}{r-1}.$$

Hence by the addition principle,

$$\begin{aligned}
 \binom{n+1}{r} &= \text{total number of ways to choose } r \text{ objects from } n+1 \text{ distinct objects} \\
 &= \binom{n}{r} + \binom{n}{r-1} \text{ (shown)}.
 \end{aligned}$$